

VELORA — Database Architecture Part 2: Inventory, Sales, Purchasing

Module: Inventory

warehouses

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	Linked branch
name	VARCHAR(100)	NOT NULL	e.g., "Gudang Utama"
code	VARCHAR(20)	NOT NULL	
type	VARCHAR(20)	DEFAULT 'main'	Enum: main, transit, defect, consignment
address	TEXT	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
is_default	TINYINT(1)	DEFAULT 0	Default warehouse for branch
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: UNIQUE(business_id, code) , (branch_id)

product_warehouse (Current Stock per Location)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE, FK→product_variants	
warehouse_id	CHAR(36)	FK→warehouses	
business_id	CHAR(36)	FK→businesses	Denormalized for query speed
quantity	DECIMAL(15,4)	DEFAULT 0	Current stock in base unit
reserved_quantity	DECIMAL(15,4)	DEFAULT 0	Reserved (pending orders)
available_quantity	DECIMAL(15,4)	GENERATED	quantity - reserved_quantity
cost_price_avg	BIGINT UNSIGNED	DEFAULT 0	Weighted average cost
last_restock_at	TIMESTAMP	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(product_id, product_variant_id, warehouse_id) , (business_id, warehouse_id)

stock_movements

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
product_id	CHAR(36)	FK→products	

Column	Type	Constraints	Description
product_variant_id	CHAR(36)	NULLABLE	
warehouse_id	CHAR(36)	FK→warehouses	
batch_id	CHAR(36)	NULLABLE, FK→batches	
reference_type	VARCHAR(100)	NOT NULL	Source model: transaction, purchase, adjustment, transfer, opname
reference_id	CHAR(36)	NOT NULL	Source model ID
movement_type	VARCHAR(20)	NOT NULL	Enum: in, out, transfer_in, transfer_out, adjustment, opname
quantity	DECIMAL(15,4)	NOT NULL	Positive for in, negative for out
unit_id	CHAR(36)	FK→units	Unit used in this movement
base_quantity	DECIMAL(15,4)	NOT NULL	Converted to base unit
cost_price	BIGINT UNSIGNED	DEFAULT 0	Cost per base unit at time
stock_before	DECIMAL(15,4)	NOT NULL	Stock before this movement
stock_after	DECIMAL(15,4)	NOT NULL	Stock after this movement
note	TEXT	NULLABLE	
performed_by	CHAR(36)	NULLABLE, FK→users	
created_at	TIMESTAMP		

Indexes: (business_id, product_id, created_at) , (reference_type, reference_id) , (warehouse_id, created_at)

 **Note**

stock_movements is **append-only** (insert-only). Never update or delete. This is the immutable ledger of all stock changes.

stock_transfers

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
transfer_number	VARCHAR(50)	UNIQUE	e.g., "TRF-20260518-001"
from_warehouse_id	CHAR(36)	FK→warehouses	
to_warehouse_id	CHAR(36)	FK→warehouses	
status	VARCHAR(20)	DEFAULT 'draft'	Enum: draft, in_transit, received, cancelled
note	TEXT	NULLABLE	
transferred_by	CHAR(36)	FK→users	
received_by	CHAR(36)	NULLABLE, FK→users	
transferred_at	TIMESTAMP	NULLABLE	
received_at	TIMESTAMP	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

stock_transfer_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
stock_transfer_id	CHAR(36)	FK→stock_transfers	
product_id	CHAR(36)	FK→products	

Column	Type	Constraints	Description
product_variant_id	CHAR(36)	NULLABLE	
quantity_sent	DECIMAL(15,4)	NOT NULL	
quantity_received	DECIMAL(15,4)	DEFAULT 0	May differ
unit_id	CHAR(36)	FK→units	
base_quantity_sent	DECIMAL(15,4)	NOT NULL	
note	TEXT	NULLABLE	
created_at	TIMESTAMP		

stock_opnames

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
warehouse_id	CHAR(36)	FK→warehouses	
opname_number	VARCHAR(50)	UNIQUE	e.g., "OPN-20260518-001"
status	VARCHAR(20)	DEFAULT 'draft'	Enum: draft, in_progress, completed, cancelled
note	TEXT	NULLABLE	
started_by	CHAR(36)	FK→users	
approved_by	CHAR(36)	NULLABLE, FK→users	
started_at	TIMESTAMP	NULLABLE	
completed_at	TIMESTAMP	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

stock_opname_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
stock_opname_id	CHAR(36)	FK→stock_opnames	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE	
system_quantity	DECIMAL(15,4)	NOT NULL	What system says
actual_quantity	DECIMAL(15,4)	NOT NULL	What was counted
difference	DECIMAL(15,4)	GENERATED	actual - system
note	TEXT	NULLABLE	Reason for difference
created_at	TIMESTAMP		

stock_adjustments

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
warehouse_id	CHAR(36)	FK→warehouses	
adjustment_number	VARCHAR(50)	UNIQUE	
reason	VARCHAR(50)	NOT NULL	Enum: damaged, lost, expired, found, production, correction, other
note	TEXT	NULLABLE	
status	VARCHAR(20)	DEFAULT 'draft'	Enum: draft, approved, rejected
adjusted_by	CHAR(36)	FK→users	
approved_by	CHAR(36)	NULLABLE, FK→users	

Column	Type	Constraints	Description
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

stock_adjustment_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
stock_adjustment_id	CHAR(36)	FK→stock_adjustments	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE	
quantity	DECIMAL(15,4)	NOT NULL	Positive = add, Negative = subtract
unit_id	CHAR(36)	FK→units	
base_quantity	DECIMAL(15,4)	NOT NULL	
cost_price	BIGINT UNSIGNED	DEFAULT 0	Cost impact
note	TEXT	NULLABLE	
created_at	TIMESTAMP		

batches

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE	
warehouse_id	CHAR(36)	FK→warehouses	

Column	Type	Constraints	Description
batch_number	VARCHAR(50)	NOT NULL	Lot/batch code
quantity	DECIMAL(15,4)	DEFAULT 0	Current qty in batch
cost_price	BIGINT UNSIGNED	DEFAULT 0	Cost per unit in this batch
manufactured_at	DATE	NULLABLE	Production date
expired_at	DATE	NULLABLE	Expiry date
status	VARCHAR(20)	DEFAULT 'active'	Enum: active, depleted, expired, recalled
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: (business_id, product_id, expired_at) , (business_id, expired_at, status)

Module: Sales

cashier_shifts

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
user_id	CHAR(36)	FK→users	Cashier
shift_number	VARCHAR(50)	UNIQUE	
opening_amount	BIGINT UNSIGNED	DEFAULT 0	Starting cash
closing_amount	BIGINT UNSIGNED	NULLABLE	Ending cash
expected_amount	BIGINT UNSIGNED	NULLABLE	System-calculated

Column	Type	Constraints	Description
difference	BIGINT	NULLABLE	closing - expected
status	VARCHAR(20)	DEFAULT 'open'	Enum: open, closed
opened_at	TIMESTAMP	NOT NULL	
closed_at	TIMESTAMP	NULLABLE	
note	TEXT	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: (business_id, branch_id, status) , (user_id, status)

payment_methods

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(50)	NOT NULL	e.g., "Cash", "BCA Transfer", "QRIS"
type	VARCHAR(30)	NOT NULL	Enum: cash, bank_transfer, ewallet, qris, card, credit, other
provider	VARCHAR(50)	NULLABLE	e.g., "BCA", "GoPay"
account_number	VARCHAR(50)	NULLABLE	
account_name	VARCHAR(100)	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
is_default	TINYINT(1)	DEFAULT 0	
sort_order	INT	DEFAULT 0	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

taxes

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(50)	NOT NULL	e.g., "PPN 11%"
rate	DECIMAL(5,2)	NOT NULL	e.g., 11.00
type	VARCHAR(20)	DEFAULT 'percentage'	Enum: percentage, fixed
is_inclusive	TINYINT(1)	DEFAULT 0	Tax included in price?
is_default	TINYINT(1)	DEFAULT 0	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

discounts

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(100)	NOT NULL	
code	VARCHAR(30)	NULLABLE	Promo code
type	VARCHAR(20)	NOT NULL	Enum: percentage, fixed, buy_x_get_y
value	DECIMAL(15,2)	NOT NULL	% or fixed amount
min_purchase	BIGINT UNSIGNED	DEFAULT 0	Minimum transaction
max_discount	BIGINT UNSIGNED	NULLABLE	Cap for percentage

Column	Type	Constraints	Description
scope	VARCHAR(20)	DEFAULT 'transaction'	Enum: transaction, product, category
applicable_ids	JSON	NULLABLE	Product/category IDs if scoped
usage_limit	INT	NULLABLE	Max total uses
usage_per_customer	INT	NULLABLE	Max per customer
used_count	INT UNSIGNED	DEFAULT 0	Current usage
starts_at	TIMESTAMP	NULLABLE	
ends_at	TIMESTAMP	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

transactions

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
warehouse_id	CHAR(36)	FK→warehouses	Stock source
cashier_shift_id	CHAR(36)	NULLABLE, FK→cashier_shifts	
customer_id	CHAR(36)	NULLABLE, FK→customers	
user_id	CHAR(36)	FK→users	Cashier/operator
transaction_number	VARCHAR(50)	UNIQUE	e.g., "TRX-20260518-00001"

Column	Type	Constraints	Description
transaction_date	DATE	NOT NULL	
transaction_type	VARCHAR(20)	DEFAULT 'sale'	Enum: sale, return, exchange
subtotal	BIGINT UNSIGNED	DEFAULT 0	Before discount/tax
discount_amount	BIGINT UNSIGNED	DEFAULT 0	Total discount
discount_id	CHAR(36)	NULLABLE, FK→discounts	Applied promo
tax_amount	BIGINT UNSIGNED	DEFAULT 0	Total tax
rounding_amount	BIGINT	DEFAULT 0	Rounding +/-
grand_total	BIGINT UNSIGNED	NOT NULL	Final amount
paid_amount	BIGINT UNSIGNED	DEFAULT 0	Amount received
change_amount	BIGINT UNSIGNED	DEFAULT 0	Change given
payment_status	VARCHAR(20)	DEFAULT 'paid'	Enum: paid, partial, unpaid, refunded
status	VARCHAR(20)	DEFAULT 'completed'	Enum: draft, completed, voided, refunded
note	TEXT	NULLABLE	
metadata	JSON	NULLABLE	Extra data
voided_at	TIMESTAMP	NULLABLE	
voided_by	CHAR(36)	NULLABLE	
void_reason	TEXT	NULLABLE	

Column	Type	Constraints	Description
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: (business_id, branch_id, transaction_date) , (business_id, transaction_number) , (customer_id) , (cashier_shift_id)

transaction_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
transaction_id	CHAR(36)	FK→transactions	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE	
batch_id	CHAR(36)	NULLABLE, FK→batches	
product_name	VARCHAR(200)	NOT NULL	Snapshot at time of sale
sku	VARCHAR(50)	NULLABLE	Snapshot
quantity	DECIMAL(15,4)	NOT NULL	
unit_id	CHAR(36)	FK→units	
base_quantity	DECIMAL(15,4)	NOT NULL	Converted to base
unit_price	BIGINT UNSIGNED	NOT NULL	Price per unit sold
cost_price	BIGINT UNSIGNED	NOT NULL	COGS per unit
subtotal	BIGINT UNSIGNED	NOT NULL	qty × unit_price

Column	Type	Constraints	Description
discount_amount	BIGINT UNSIGNED	DEFAULT 0	Item-level discount
tax_amount	BIGINT UNSIGNED	DEFAULT 0	Item-level tax
total	BIGINT UNSIGNED	NOT NULL	Final item amount
note	TEXT	NULLABLE	
created_at	TIMESTAMP		

transaction_payments (Split payment support)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
transaction_id	CHAR(36)	FK→transactions	
payment_method_id	CHAR(36)	FK→payment_methods	
amount	BIGINT UNSIGNED	NOT NULL	
reference	VARCHAR(100)	NULLABLE	Card/transfer ref
status	VARCHAR(20)	DEFAULT 'completed'	
created_at	TIMESTAMP		

refunds

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
transaction_id	CHAR(36)	FK→transactions	Original transaction
refund_number	VARCHAR(50)	UNIQUE	

Column	Type	Constraints	Description
refund_type	VARCHAR(20)	NOT NULL	Enum: full, partial, exchange
total_amount	BIGINT UNSIGNED	NOT NULL	
refund_method	VARCHAR(30)	NOT NULL	cash, credit, store_credit
reason	TEXT	NULLABLE	
status	VARCHAR(20)	DEFAULT 'completed'	Enum: pending, approved, completed, rejected
refunded_by	CHAR(36)	FK→users	
approved_by	CHAR(36)	NULLABLE, FK→users	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

refund_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
refund_id	CHAR(36)	FK→refunds	
transaction_item_id	CHAR(36)	FK→transaction_items	
quantity	DECIMAL(15,4)	NOT NULL	
amount	BIGINT UNSIGNED	NOT NULL	
return_to_stock	TINYINT(1)	DEFAULT 1	Re-stock the item?
condition	VARCHAR(20)	DEFAULT 'good'	Enum: good, damaged, expired
note	TEXT	NULLABLE	
created_at	TIMESTAMP		

Module: Purchasing

suppliers

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(150)	NOT NULL	
code	VARCHAR(30)	NULLABLE	
contact_person	VARCHAR(100)	NULLABLE	
phone	VARCHAR(20)	NULLABLE	
email	VARCHAR(150)	NULLABLE	
address	TEXT	NULLABLE	
city	VARCHAR(100)	NULLABLE	
tax_id	VARCHAR(50)	NULLABLE	Supplier NPWP
payment_terms	INT	DEFAULT 0	Days for payment (0 = COD)
is_active	TINYINT(1)	DEFAULT 1	
notes	TEXT	NULLABLE	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

purchases

Column	Type	Constraints	Description
id	CHAR(36)	PK	

Column	Type	Constraints	Description
business_id	CHAR(36)	FK→businesses	
branch_id	CHAR(36)	FK→branches	
warehouse_id	CHAR(36)	FK→warehouses	Receiving warehouse
supplier_id	CHAR(36)	FK→suppliers	
user_id	CHAR(36)	FK→users	Creator
purchase_number	VARCHAR(50)	UNIQUE	e.g., "PO-20260518-001"
purchase_date	DATE	NOT NULL	
expected_date	DATE	NULLABLE	Expected delivery
received_date	DATE	NULLABLE	Actual delivery
subtotal	BIGINT UNSIGNED	DEFAULT 0	
discount_amount	BIGINT UNSIGNED	DEFAULT 0	
tax_amount	BIGINT UNSIGNED	DEFAULT 0	
shipping_cost	BIGINT UNSIGNED	DEFAULT 0	
grand_total	BIGINT UNSIGNED	NOT NULL	
paid_amount	BIGINT UNSIGNED	DEFAULT 0	
payment_status	VARCHAR(20)	DEFAULT 'unpaid'	Enum: unpaid, partial, paid
status	VARCHAR(20)	DEFAULT 'draft'	Enum: draft, ordered, partial_received, received, completed, cancelled
note	TEXT	NULLABLE	

Column	Type	Constraints	Description
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

purchase_items

Column	Type	Constraints	Description
id	CHAR(36)	PK	
purchase_id	CHAR(36)	FK→purchases	
product_id	CHAR(36)	FK→products	
product_variant_id	CHAR(36)	NULLABLE	
quantity_ordered	DECIMAL(15,4)	NOT NULL	
quantity_received	DECIMAL(15,4)	DEFAULT 0	
unit_id	CHAR(36)	FK→units	Purchase unit
base_quantity_ordered	DECIMAL(15,4)	NOT NULL	In base unit
base_quantity_received	DECIMAL(15,4)	DEFAULT 0	
unit_price	BIGINT UNSIGNED	NOT NULL	Price per purchase unit
discount_amount	BIGINT UNSIGNED	DEFAULT 0	
tax_amount	BIGINT UNSIGNED	DEFAULT 0	
total	BIGINT UNSIGNED	NOT NULL	
batch_number	VARCHAR(50)	NULLABLE	Lot for batch tracking
expired_at	DATE	NULLABLE	
note	TEXT	NULLABLE	
created_at	TIMESTAMP		

supplier_debts

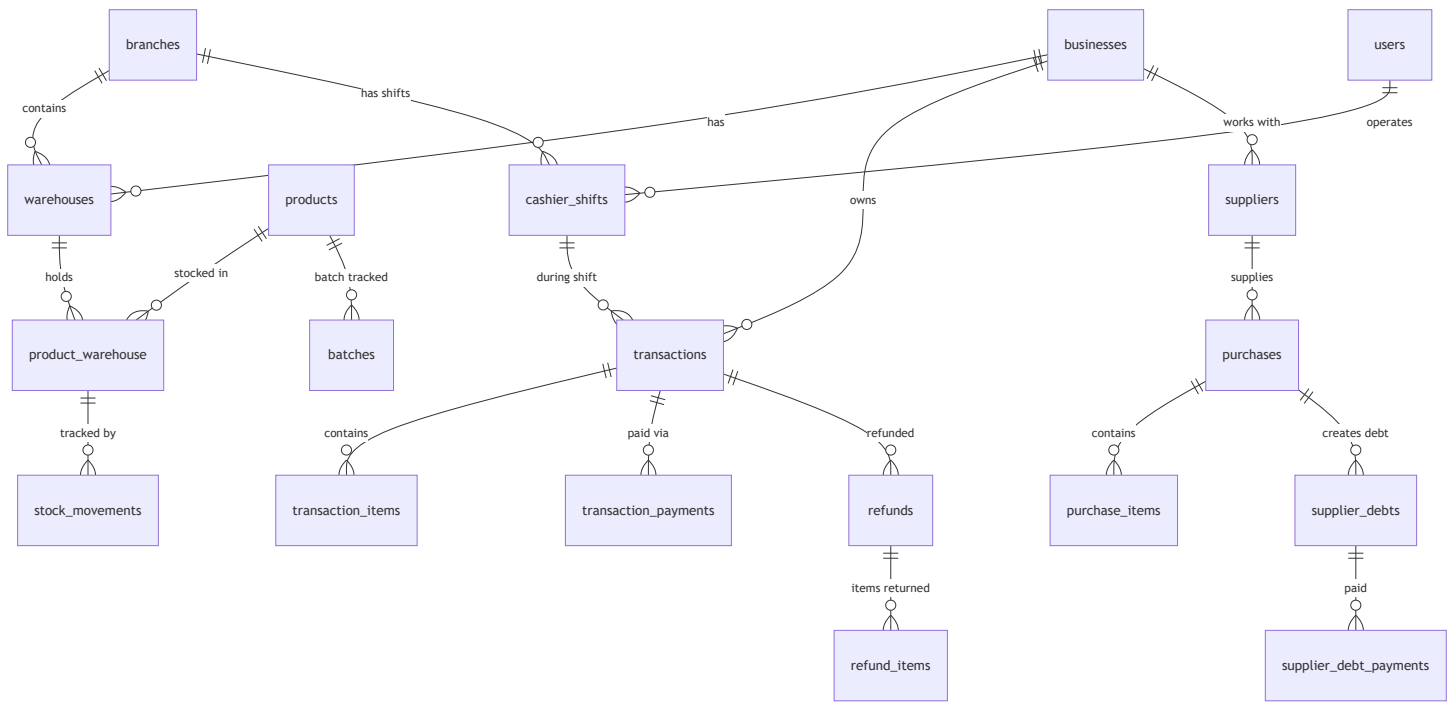
Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
supplier_id	CHAR(36)	FK→suppliers	
purchase_id	CHAR(36)	FK→purchases	
total_amount	BIGINT UNSIGNED	NOT NULL	Total debt
paid_amount	BIGINT UNSIGNED	DEFAULT 0	Amount paid so far
remaining_amount	BIGINT UNSIGNED	GENERATED	total - paid
due_date	DATE	NOT NULL	
status	VARCHAR(20)	DEFAULT 'unpaid'	Enum: unpaid, partial, paid, overdue
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

supplier_debt_payments

Column	Type	Constraints	Description
id	CHAR(36)	PK	
supplier_debt_id	CHAR(36)	FK→supplier_debts	
amount	BIGINT UNSIGNED	NOT NULL	
payment_method_id	CHAR(36)	FK→payment_methods	
reference	VARCHAR(100)	NULLABLE	
note	TEXT	NULLABLE	

Column	Type	Constraints	Description
paid_by	CHAR(36)	FK→users	
paid_at	TIMESTAMP	NOT NULL	
created_at	TIMESTAMP		

Relationship Diagram — Inventory, Sales, Purchasing



Continued in [Part 3 — CRM, Finance, Settings](file:///C:/Users/YP/.gemini/antigravity/brain/7880df49-c88e-4c8f-82bc-eb5a42cebb61/database_architecture_part3.md)